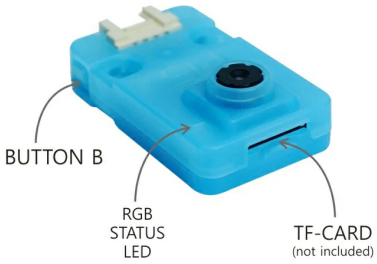
UnitV-OV2640

SKU:U078







Description

UnitV-OV2640 is an AI vision processing camera unit equipped with the Kendryte K210, integrating a dual-core 64-bit RISC-V CPU and a neural network processor edge computing system-on-chip. The **UnitV-OV2640** AI camera is very compact, making it suitable for embedding into various devices. It has machine vision processing capabilities, supports multiple image recognition functions (such as real-time acquisition of the size and coordinates of detected objects, and real-time acquisition of the types of detected objects), and can perform convolutional neural network calculations under low power consumption. Therefore, **UnitV-OV2640** is an excellent zero-threshold machine vision embedded solution. It supports the MicroPython development environment, making your project development code more concise. Equipped with the OV2640 2-megapixel image sensor, it is an ideal choice for machine vision projects. The body features two programmable buttons and a front-facing RGB LED indicator for easy status display. The bottom provides a HY2.0-4P compatible interface and a TYPE-C interface for connecting to the main control device. It supports TF card expansion for easier access to related materials and model files.

Features

- Dual-core 64-bit RISC-V RV64IMAFDC (RV64GC) CPU / 400MHz (Normal)
- Double-precision FPU
- 8MiB 64-bit on-chip SRAM
- Neural Network Processor (KPU) / 0.8Tops
- Programmable I/O Array (FPIOA)
- AES, SHA256 Accelerator
- Direct Memory Access Controller (DMAC)
- Supports MicroPython
- Firmware encryption support
- On-board hardware resources:
  - Flash: 16M
  - Camera: OV2640
  - Buttons: 2
  - Status LED: WS2812 LED
  - Expansion card interface: TF card/microSD
  - Interface: HY2.0/compatible GROVE

Includes

- 1 x UnitV-OV2640
- 1 x HY2.0-4P Grove cable (20cm)
- 1 x USB Type-C cable (1m)

Applications

- Object detection/classification
- Real-time acquisition of target size and coordinates
- Real-time acquisition of detected target types
- Shape recognition
- Video recording

Specifications

Specification	Parameters
Kendryte K210	Dual-core 64-bit RISC-V RV64IMAFDC (RV64GC) CPU / 400MHz (Normal)
SRAM	8Mbit
Flash	16M
Input voltage	5V @ 500mA
KPU Neural Network Size	5.5MiB-5.9MiB
Interface	Type-C x 1, HY2.0-4P (I2C+I/O+UART) x 1
RGB LED	WS2812 x 1
Buttons	Custom buttons x 2
Camera	OV2640
FOV	65°
External storage	TF Card/microSD
Net weight	8g
Gross Weight	45g
Product Size	40 x 24 x 13mm
Package Size	70 x 50 x 30mm
Casing material	Plastic (PC)

Learn

About KENDRYTE K210

Kendryte K210 is a system-on-chip (SoC) with integrated machine vision capabilities. Using TSMC's ultra-low-power 28nm advanced process, it features a dual-core 64-bit processor with excellent power efficiency, stability, and reliability. This solution aims for zero-threshold development, enabling rapid deployment in user products with AI applications.

- Equipped with machine vision capabilities
- Better low-power vision processing speed and accuracy
- Equipped with a convolutional neural network hardware accelerator KPU for high-performance convolutional neural network operations
- TSMC 28nm advanced process, temperature range -40°C to 125°C, stable and reliable
- Supports firmware encryption, difficult to crack using ordinary methods

- Unique programmable IO array, making product design more flexible
- Low voltage, lower power consumption compared to systems with the same processing power
- 3.3V/1.8V dual voltage support, no level conversion required, cost-saving

This product features a dual-core 64-bit high-performance low-power CPU based on RISC-V ISA, with the following characteristics:

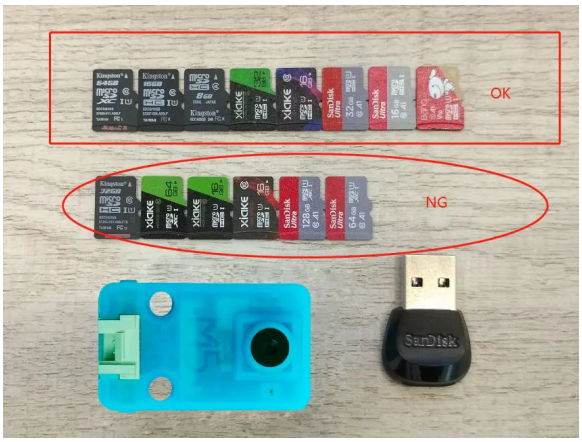
- Core count: Dual-core processor
- Processor width: 64-bit CPU 400MHz
- Nominal frequency: 400MHz
- Instruction set extension: IMAFDC
- Floating-point unit (FPU): Double precision
- Platform interrupt management: PLIC
- Local interrupt management: CLINT
- Instruction cache: 32KiB x 2
- Data cache: 32KiB x 2
- On-chip SRAM: 8MiB

About OV2640

- Supported output formats (8-bit):
  - YUV (422/420) /YCbCr422
  - RGB565/555
  - 8-bit compressed data
  - 8-/10-bit Raw RGB data
- Scanning mode: Progressive
- Camera specifications:
  - CCD size: 1/4 inch
  - Field of view: 65°
  - Maximum pixels: 2M
  - Sensor optimal resolution: 1600 x 1200

SD Card Test

Unit V currently cannot recognize all types of microSD cards. We have tested some common microSD cards, and the results are as follows.



Brand	Memory	Type	Transfer Speed	Partition Format	Test Result
Kingston	8G	HC	Class4	FAT32	OK
Kingston	16G	HC	Class10	FAT32	OK
Kingston	32G	HC	Class10	FAT32	NO
Kingston	64G	XC	Class10	exFAT	OK
SanDisk	16G	HC	Class10	FAT32	OK
SanDisk	32G	HC	Class10	FAT32	OK
SanDisk	64G	XC	Class10	/	NO
SanDisk	128G	XC	Class10	/	NO
XIAKE	16G	HC	Class10	FAT32	OK (Purple)
XIAKE	32G	HC	Class10	FAT32	OK
XIAKE	64G	XC	Class10	/	NO
TURYE	32G	HC	Class10	/	NO

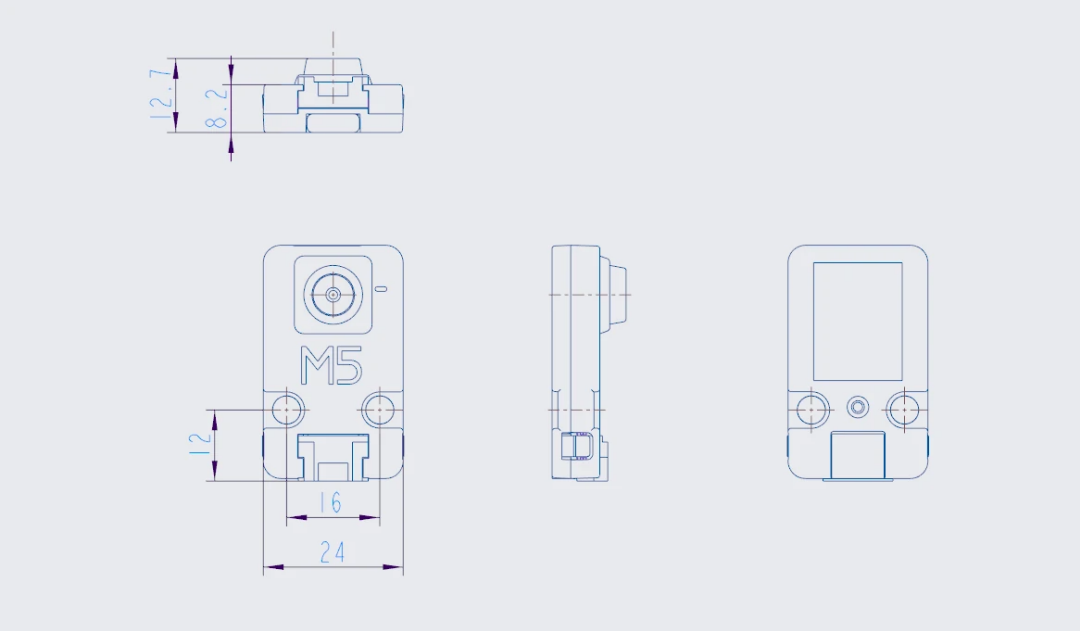
PinMap

UnitV-OV2640

UnitV	G8	G19	G18	G34, G35
Hardware	RGB LED	Button A	Button B	
HY2.0-4P				Interface



Model Size



Datasheets

- K210 Datasheet

Softwares

Quick Start

Choose the development platform you want to use and check the corresponding tutorial & quick start.

- UnitV-OV2640 V-Function Guide
- UnitV-OV2640 V-Training Guide
- UnitV-OV2640 Maixpy Guide

UiFlow1

- UnitV-OV2640 UiFlow1 Docs

UiFlow2

- Coming soon...

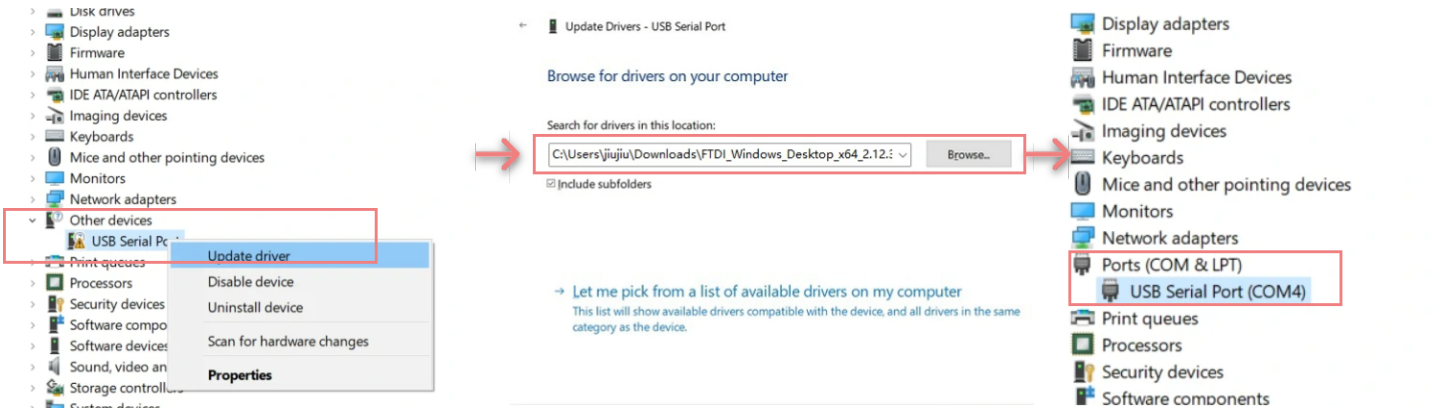
USB Driver

UnitV may not work without a driver on some systems. You can resolve this issue by manually installing the FTDI VCP driver. Follow these steps:

1. Select and download the driver file matching your operating system from the list below, then extract it. You can also visit the FTDI Chip official website to get additional FTDI driver versions.

Driver name	Download link
Windows_Desktop_x64	Download
Windows_Desktop_x86_32	Download
Windows_Desktop_ARM	Download
MacOS_Intel	Download
MacOS_ARM64	Download

2. Connect the device to your PC, open Device Manager, and install the FTDI driver (the screenshot below uses Windows 10 as an example). (Note: On some systems, you may need to install the driver twice for it to take effect. The unrecognized device is usually named `M5Stack` or `USB Serial`.)



Other

- UnitV-OV2640 with RoverC Track Ball Example

Video

unitV.mp4

Product Comparison

To compare information on the UnitV series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.